CTU: Mechanics Lectures Compilation

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Chapter 1

Mathematical Tools needed for this lecture

Information about this chapter blah blah blah

1.1 Scalars

Scalars are quantities that are fully described by a magnitude (or numerical value) alone.

Examples of scalars in physics include mass, temperature, and energy:

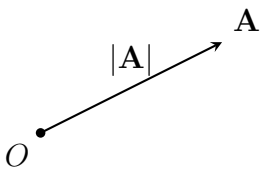
- 1 kilogram (mass)
- 20 degrees Celcius (temperature)
- 100 Joules (energgy)

Scalars can be *added, substracted, multiplied* and *divided* using the standard rules of arithmetic.

Scalars can only be added or substracted when they share the same physical dimension. For example, $1kg + 2kg = 3kg$ is valid but, $1kg + 20C$ is meaningless. However, scalars of different dimensions can be multiplied or divided, yielding a new unit.

1.2 Vectors

Vectors are quantities that have both a magnitude and a direction. They can be represented graphically as arrows, where the length of the arrow represents

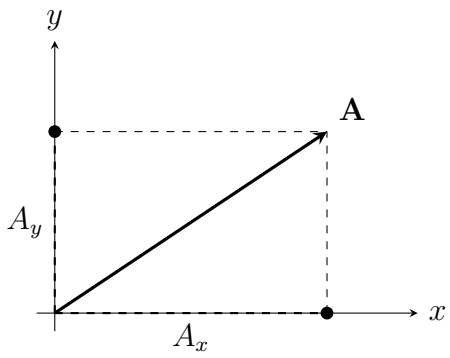


the magnitude and the direction of the arrow indicates the direction of the vector.

We will represent vectors using boldface letters in these notes, such as $\mathbf{A}$, or with an arrow above the letter on the blackboard, such as $\vec{A}$. The magnitude of a vector $\mathbf{A}$ is denoted by $|\mathbf{A}|$ or A.

1.2.1 Components of Vectors

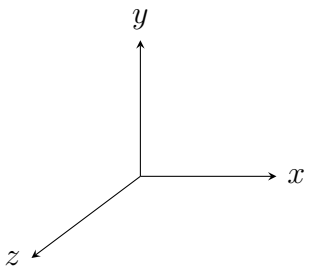
A vector can be broken down into its components along the coordinate axes. For example, a vector $\mathbf{A}$ in two-dimensional space can be expressed in terms of its components along the x-axis and y-axis by two numbers A_x and A_y as it can be seen in the figure below.



It is conventional to express a vector in terms of a column vector of its components along the coordinate axes. For example, a vector $\mathbf{A}$ in two-dimensional space can be expressed as:

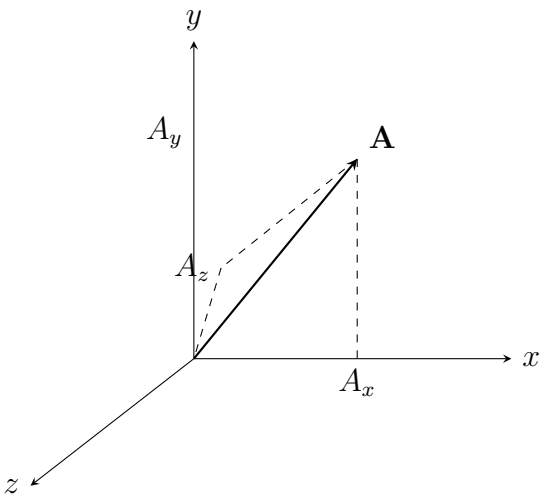
$$\mathbf{A} = \begin{bmatrix} A_x \\ A_y \end{bmatrix}.$$

The real world is three-dimensional, and when it comes to draw vectors in three-dimensional space on a paper, we need to use a three coordinate system. The way we draw a three-dimensional coordinate system on a paper is shown in the figure below:



One can think of these axes as the edges of a cube. The x-axis points to the right, the y-axis points up, and the z-axis points out of the page towards you. The positive direction of each axis is indicated by an arrowhead. A vector is in three-dimensional space then can be expressed as:

$$\mathbf{A} = \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}.$$



The magnitude of a vector $\mathbf{A}$ in three-dimensional space can therefore be calculated using the Pythagorean theorem:

$$|\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}.$$

1.2.2 Unit Vectors

A *unit vector* is a vector with a magnitude of 1. It is used to indicate direction without specifying magnitude. A unit vector can be obtained by dividing a

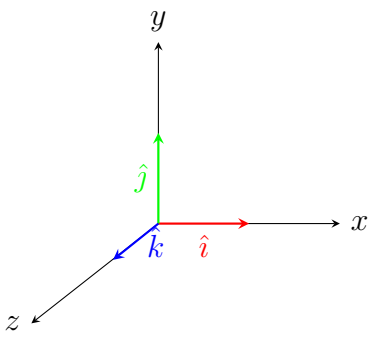
vector by its magnitude:

$$\mathbf{a} = \frac{\mathbf{A}}{|\mathbf{A}|} = \begin{bmatrix} \frac{A_x}{|\mathbf{A}|} \\ \frac{A_y}{|\mathbf{A}|} \\ \frac{A_z}{|\mathbf{A}|} \end{bmatrix}.$$

We can define *special unit vectors* along the coordinate axes. In three-dimensional space, we define the unit vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ as follows:

$$\mathbf{i} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

On the paper, we use hat notation for unit vectors, such as $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$. The unit vectors are shown in the figure below:



Once we define the unit vectors, we can express any vector in three-dimensional space as a linear combination of the unit vectors:

$$\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}.$$

1.3 Operations on Vectors

Like scalars, vectors can be added, substracted, multiplied, and divided.

1.3.1 Addition and Substraction

1.3.2 Scalar Multiplication

Scalar multiplication only changes the magnitude of a vector, not its direction. If the scalar is negative, it also reverses the direction of the vector:

$$c\mathbf{A} = cA_x\mathbf{i} + cA_y\mathbf{j} + cA_z\mathbf{k}.$$

1.3.3 Dot Product

The *dot product* of two vectors is defined as the product of their magnitudes and the cosine of the angle between them:

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos\theta = A_x B_x + A_y B_y + A_z B_z.$$

The dot product is useful for finding the component of one vector along another and for calculating angles between vectors.

1.3.4 Cross Product

The *cross product* of two vectors produces a vector perpendicular to both:

$$\mathbf{A} \times \mathbf{B} = |\mathbf{A}||\mathbf{B}| \sin\theta\, \hat{\mathbf{n}},$$

where $\hat{\mathbf{n}}$ is determined by the right-hand rule. In three dimensions,

$$\begin{aligned}
\mathbf{A} \times \mathbf{B} &= (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \times (B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}) \\
&= A_x B_x(\mathbf{i} \times \mathbf{i}) + A_x B_y(\mathbf{i} \times \mathbf{j}) + A_x B_z(\mathbf{i} \times \mathbf{k}) \\
&\quad + A_y B_x(\mathbf{j} \times \mathbf{i}) + A_y B_y(\mathbf{j} \times \mathbf{j}) + A_y B_z(\mathbf{j} \times \mathbf{k}) \\
&\quad + A_z B_x(\mathbf{k} \times \mathbf{i}) + A_z B_y(\mathbf{k} \times \mathbf{j}) + A_z B_z(\mathbf{k} \times \mathbf{k}) \\
&= (A_y B_z - A_z B_y)\mathbf{i} + (A_z B_x - A_x B_z)\mathbf{j} + (A_x B_y - A_y B_x)\mathbf{k}.
\end{aligned}$$

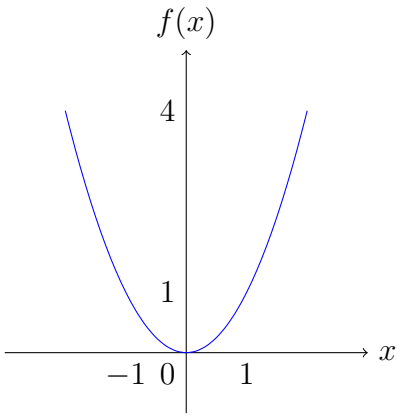
1.4 Basic Calculus

1.4.1 Function

A *function* is a rule that assigns a unique output to each input. For example, the function $f(x) = x^2$ assigns the square of x to each value of x. The value x is called the *parameter* of function f and the value $f(x)$ is called the *output* of the function for that parameter. It is conventional to draw a figure to represent a function to have a better understanding of its behavior, where the horizontal axis represents the input values and the vertical axis represents the output values. For example, the function $f(x) = x^2$ can be represented as follows:

1.4.2 Derivative

For a function $f(x)$, its *derivative* describes the rate at which the output of the function changes with respect to its input. It is defined as the ratio of



the change in the output to the change in the input as the change in the input approaches zero. This is depicted in the following figure:

Figure 1.1: Geometrical interpretation of the derivative.

The derivative of a function $f(x)$ with respect to x is denoted by $\frac{df}{dx}$ or $f'(x)$. It is defined as the limit of the average rate of change of the function as the interval approaches zero:

$$\frac{df}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}.$$

Geometrically, the derivative $f'(x)$ gives the slope of the tangent to the graph of $f(x)$ at a given point.

1.4.3 Different Notations for Derivatives

There are several common ways to denote the derivative of a function:

- $\frac{df}{dx}$: This is the *Leibniz notation*. It explicitly indicates that the derivative is taken with respect to x.
- $f'(x)$: This is the *Lagrange notation*, also called *prime notation*. It is a concise notation commonly used when working with functions.
- $\dot{f}$: This is the *Newton notation*, also called *dot notation*. It denotes differentiation with respect to time and is particularly common in physics and mechanics. For example,

$$\dot{f} = \frac{df}{dt}.$$

1.4.4 Derivatives of Common Functions

The derivatives of some common functions are as follows:

$$\begin{aligned}
\frac{d}{dx}(c) &= 0, \\
\frac{d}{dx}(x^n) &= nx^{n-1}, \\
\frac{d}{dx}(e^x) &= e^x, \\
\frac{d}{dx}(\ln x) &= \frac{1}{x}, \\
\frac{d}{dx}(\sin x) &= \cos x, \\
\frac{d}{dx}(\cos x) &= -\sin x, \\
\frac{d}{dx}(\tan x) &= \sec^2 x, \\
\frac{d}{dx}(\arcsin x) &= \frac{1}{\sqrt{1-x^2}}, \\
\frac{d}{dx}(\arccos x) &= -\frac{1}{\sqrt{1-x^2}}, \\
\frac{d}{dx}(\arctan x) &= \frac{1}{1+x^2}.
\end{aligned}$$

Some useful rules for differentiating functions $f(x)$ and $g(x)$ are

$$\begin{aligned}
\frac{d}{dx}[f(x)+g(x)] &= f'(x)+g'(x), \\
\frac{d}{dx}[cf(x)] &= cf'(x), \\
\frac{d}{dx}[f(x)g(x)] &= f'(x)g(x)+f(x)g'(x), \\
\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] &= \frac{f'(x)g(x)-f(x)g'(x)}{[g(x)]^2}.
\end{aligned}$$

For a composite function $f(g(x))$, the chain rule gives

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x).$$

1.4.5 Multivariable Functions and Partial Derivatives

$f = f(x, y, z, \ldots)$ is a *multivariable function* if it depends on more than one variable. For example, the surface $f(x, y) = x^2 + y^2$ is shown below.

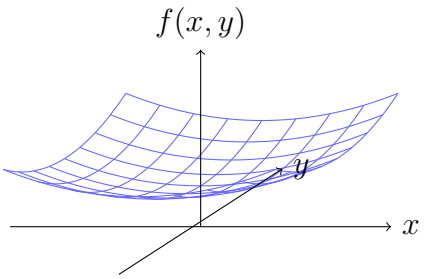


Figure 1.2: The surface $f(x, y) = x^2 + y^2$.

The *partial derivative* of a multivariable function with respect to one of its variables is the derivative of the function while treating all other variables as constants. For example, the partial derivative of $f(x, y)$ with respect to x is denoted by $\frac{\partial f}{\partial x}$ and is defined as

$$\frac{\partial f}{\partial x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}.$$

For example, for the function $f(x, y) = x^2 + y^2$, the partial derivatives are

$$\frac{\partial f}{\partial x} = 2x, \quad \frac{\partial f}{\partial y} = 2y.$$

For example, the partial derivative of $g(x, y) = x^2 + xy^2$ with respect to x is

$$\frac{\partial g}{\partial x} = 2x + y^2,$$

and the partial derivative with respect to y is

$$\frac{\partial g}{\partial y} = 2xy.$$

1.4.6 Differential and Total Differentials

The smallest change in the value of a function when we change all of its input variables is called the *differential* of the function. For a function $f = f(x, y)$, the differential df is given by

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy,$$

where dx and dy are the small changes in the input variables x and y, respectively. The differential provides an approximation of the change in the function's value based on the changes in its input variables. In general, for a function $f = f(x_1, x_2, \ldots, x_n)$, the differential is given by

$$df = \frac{\partial f}{\partial x_1} dx_1 + \frac{\partial f}{\partial x_2} dx_2 + \cdots + \frac{\partial f}{\partial x_n} dx_n.$$

In a compact form, the differential can be expressed as

$$df = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} dx_i.$$

1.4.7 Total derivative

The *total derivative* of a function $f = f(x_1, x_2, \ldots, x_n, t)$ and $x_i(t) = x_i$ where $i = 1, 2, \ldots, n$ is given by

$$\frac{df}{dt} = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i} \frac{dx_i}{dt} + \frac{\partial f}{\partial t}.$$

1.4.8 Maxima and Minima of Functions

If we have a function $f(x)$, we can find its maxima and minima by taking its derivative and setting it equal to zero:

$$f'(x) = 0.$$

We solve this equation for x.

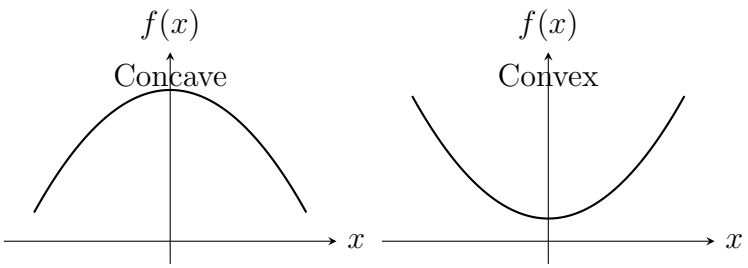
1.4.9 Concave and Convex Functions

A function $f(x)$ is *concave* if its second derivative is negative for all x in its domain:

$$f''(x) < 0.$$

A function $f(x)$ is *convex* if its second derivative is positive for all x in its domain:

$$f''(x) > 0.$$



1.4.10 Antiderivatives

The *antiderivative* of a function $f(x)$ is a function $F(x)$ whose derivative is $f(x)$:

$$F'(x) = f(x).$$

For example, since

$$\frac{d}{dx}(x^3) = 3x^2,$$

x^3 is an antiderivative of $3x^2$.

However, x^3+C is also an antiderivative, where C is an arbitrary constant, because

$$\frac{d}{dx}(x^3 + C) = 3x^2.$$

Therefore, an antiderivative is not unique: any two antiderivatives of the same function differ by a constant.

1.4.11 Indefinite Integrals

The *indefinite integral* represents the family of all antiderivatives of a function:

$$\int f(x)\,dx = F(x) + C,$$

where

$$F'(x) = f(x).$$

For example,

$$\int 3x^2\,dx = x^3 + C.$$

Some common indefinite integrals are

$$\begin{aligned}
\int c\,dx &= cx + C, \\
\int x^n\,dx &= \frac{x^{n+1}}{n+1} + C, \qquad n \neq -1, \\
\int e^x\,dx &= e^x + C, \\
\int \frac{1}{x}\,dx &= \ln|x| + C, \\
\int \cos x\,dx &= \sin x + C, \\
\int \sin x\,dx &= -\cos x + C.
\end{aligned}$$

The constant C appears because the derivative of a constant is zero.

1.4.12 Definite Integrals

A *definite integral* has specified lower and upper limits:

$$\int_a^b f(x)\,dx.$$

Unlike an indefinite integral, a definite integral has a numerical value. It represents the signed area between the graph of $f(x)$ and the x-axis from $x = a$ to $x = b$. Regions where $f(x) < 0$ contribute negatively to the integral.

If $F(x)$ is an antiderivative of $f(x)$, then the Fundamental Theorem of Calculus gives

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

For example,

$$\int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

Thus, differentiation and integration are inverse operations. Differentiation describes the instantaneous rate of change of a function, while integration can be used to accumulate a quantity or calculate the signed area under a curve.

1.5 Taylor Series

The *Taylor series* of a function $f(x)$ about a point x_0 is an infinite sum of terms that approximates the function near that point. The reason we use Taylor series is that it allows us to approximate complicated functions with polynomials, which are easier to work with.

Theorem: An *analytic function* can be written as a convergent power series. An analytic function is a function that can be infinitely differentiable. The examples of analytic functions include polynomials, exponential functions, and trigonometric functions. Therefore, if $f(x)$ is analytic about x_0, then

$$f(x) = c_0 + c_1(x - x_0) + c_2(x - x_0)^2 + c_3(x - x_0)^3 + \cdots$$

To determine the coefficients c_n, let us consider the derivatives of $f(x)$:

$$\begin{aligned}
f'(x) &= c_1 + 2c_2(x - x_0) + 3c_3(x - x_0)^2 + 4c_4(x - x_0)^3 + \cdots \\
f''(x) &= 2c_2 + 3 \cdot 2c_3(x - x_0) + 4 \cdot 3c_4(x - x_0)^2 + \cdots \\
f'''(x) &= 3 \cdot 2c_3 + 4 \cdot 3 \cdot 2c_4(x - x_0) + \cdots \\
f^{(4)}(x) &= 4 \cdot 3 \cdot 2c_4 + \cdots
\end{aligned}$$

Therefore, by evaluating these derivatives at $x = x_0$, we can find the coefficients c_n:

$$\begin{aligned}
f(x_0) = c_0 &\implies c_0 = f(x_0) \\
f'(x_0) = c_1 &\implies c_1 = f'(x_0) \\
f''(x_0) = 2c_2 &\implies c_2 = \frac{f''(x_0)}{2} \\
f'''(x_0) = 3 \cdot 2c_3 &\implies c_3 = \frac{f'''(x_0)}{3 \cdot 2} \\
f^{(4)}(x_0) = 4 \cdot 3 \cdot 2c_4 &\implies c_4 = \frac{f^{(4)}(x_0)}{4 \cdot 3 \cdot 2}
\end{aligned}$$

$$f(x) = c_0 + f'(x_0)(x - x_0) + \frac{f''(x_0)}{2}(x - x_0)^2 + \frac{f'''(x_0)}{3 \cdot 2}(x - x_0)^3 + \frac{f^{(4)}(x_0)}{4 \cdot 3 \cdot 2}(x - x_0)^4 + \cdots$$

This is called the Taylor series of $f(x)$ about x_0. The Taylor series can be expressed in a more compact form as follows:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$$

1.6 Worked Examples

1.6.1 Example 1: Scalars, Units, and the Anatomy of a Vector

(a) A student writes $m = 2\,\text{kg} + 20\,^\circ\text{C}$. Why is this meaningless? Then, for a metal block of mass $m = 2.7\,\text{kg}$ and volume $V = 1.0 \times 10^{-3}\,\text{m}^3$, compute the density.

(b) Show by dimensional analysis that $T = 2\pi\sqrt{L/g}$ can be a time.

(c) For $\mathbf{A} = 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$, find $|\mathbf{A}|$, the unit vector $\hat{a}$, and the angle α between $\mathbf{A}$ and the x-axis.

Solution

(a) Addition requires the same physical dimension; mass and temperature have different dimensions, so the sum has no meaning. Multiplication and division are always allowed and create a new unit:

$$\rho = \frac{m}{V} = \frac{2.7\,\text{kg}}{1.0 \times 10^{-3}\,\text{m}^3} = 2700\ \text{kg/m}^3.$$

(b)

$$\left[\sqrt{L/g}\right] = \sqrt{\frac{\text{m}}{\text{m/s}^2}} = \sqrt{\text{s}^2} = \text{s},$$

and 2π is a dimensionless scalar.

(c)

$$|\mathbf{A}| = \sqrt{2^2 + 3^2 + 6^2} = \sqrt{49} = 7,$$

$$\hat{a} = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{2}{7}\hat{\imath} + \frac{3}{7}\hat{\jmath} + \frac{6}{7}\hat{k}, \qquad |\hat{a}| = \frac{1}{7}\sqrt{4 + 9 + 36} = 1.$$

The direction cosine with the x-axis is

$$\cos\alpha = \frac{A_x}{|\mathbf{A}|} = \frac{2}{7} \quad \Longrightarrow \quad \alpha \approx 73.4^\circ.$$

Note also that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{4+9+36}{49} = 1$.

Board tip: draw the three-dimensional axes and the projections A_x, A_y, A_z while doing part (c).

1.6.2 Example 2: Vector Algebra and Force Equilibrium

Let $\mathbf{A} = 3\hat{\imath} - \hat{\jmath} + 2\hat{k}$ and $\mathbf{B} = -\hat{\imath} + 4\hat{\jmath} + 2\hat{k}$.

(a) Find $\mathbf{A} + \mathbf{B}$, $\mathbf{A} - \mathbf{B}$, $|\mathbf{A} + \mathbf{B}|$ and $2\mathbf{A} - 3\mathbf{B}$.

(b) A hiker walks 4 km east, then 3 km north. Find the displacement (magnitude and direction) and compare it with the distance travelled, which is a scalar.

(c) Two forces act on a particle: $\mathbf{F}_1 = (5\hat{\imath} + 2\hat{\jmath})$ N and $\mathbf{F}_2 = (-3\hat{\imath} + 6\hat{\jmath})$ N. What third force $\mathbf{F}_3$ keeps the particle in equilibrium?

Solution

(a)

$$\mathbf{A} + \mathbf{B} = 2\hat{\imath} + 3\hat{\jmath} + 4\hat{k}, \qquad |\mathbf{A} + \mathbf{B}| = \sqrt{4 + 9 + 16} = \sqrt{29} \approx 5.39,$$

$$\mathbf{A} - \mathbf{B} = 4\hat{\imath} - 5\hat{\jmath} + 0\hat{k},$$

$$2\mathbf{A} = 6\hat{\imath} - 2\hat{\jmath} + 4\hat{k}, \qquad 3\mathbf{B} = -3\hat{\imath} + 12\hat{\jmath} + 6\hat{k},$$

$$2\mathbf{A} - 3\mathbf{B} = 9\hat{\imath} - 14\hat{\jmath} - 2\hat{k}.$$

Note that $|\mathbf{A} + \mathbf{B}| \neq |\mathbf{A}| + |\mathbf{B}|$, since $\sqrt{29} \approx 5.39$ while $\sqrt{14} + \sqrt{21} \approx 8.3$.

(b) The displacement is $\mathbf{d} = 4\hat{\imath} + 3\hat{\jmath}$ km, so

$$|\mathbf{d}| = \sqrt{16 + 9} = 5 \text{ km}, \qquad \theta = \arctan\frac{3}{4} = 37^\circ \text{ north of east.}$$

The distance travelled is 7 km. It is a scalar, it has no direction, and it is different from the magnitude of the displacement.

(c) Equilibrium requires $\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 = 0$, so

$$\mathbf{F}_3 = -(\mathbf{F}_1 + \mathbf{F}_2) = -(2\hat{\imath} + 8\hat{\jmath}) = (-2\hat{\imath} - 8\hat{\jmath}) \text{ N},$$

$$|\mathbf{F}_3| = \sqrt{4 + 64} = 2\sqrt{17} \approx 8.2 \text{ N},$$

pointing 76° below the $-x$ axis, i.e. in the third quadrant.

Board tip: do part (b) graphically first with the triangle or parallelogram rule, then analytically.

1.6.3 Example 3: Dot Product, Angle, Projection and Work

Let $\mathbf{A} = 2\hat{\imath} + 3\hat{\jmath} - \hat{k}$ and $\mathbf{B} = \hat{\imath} - 2\hat{\jmath} + 2\hat{k}$.

(a) Find $\mathbf{A} \cdot \mathbf{B}$ and the angle θ between them.

(b) Find the scalar projection of $\mathbf{A}$ on $\mathbf{B}$, and the vector projection.

(c) Find c such that $\mathbf{A}$ is perpendicular to $\mathbf{C} = \hat{\imath} + c\,\hat{\jmath}$.

(d) A force $\mathbf{F} = (3\hat{\imath}+4\hat{\jmath})$ N acts while the object undergoes a displacement $\mathbf{d} = (2\hat{\imath} - \hat{\jmath})$ m. Find the work $W = \mathbf{F} \cdot \mathbf{d}$.

(e) Derive the law of cosines from $|\mathbf{A} - \mathbf{B}|^2$.

Solution

(a)

$$\mathbf{A} \cdot \mathbf{B} = (2)(1) + (3)(-2) + (-1)(2) = 2 - 6 - 2 = -6,$$

$$|\mathbf{A}| = \sqrt{4 + 9 + 1} = \sqrt{14}, \qquad |\mathbf{B}| = \sqrt{1 + 4 + 4} = 3,$$

$$\cos\theta = \frac{-6}{3\sqrt{14}} = \frac{-2}{\sqrt{14}} \approx -0.535 \quad \Longrightarrow \quad \theta \approx 122^\circ.$$

A negative dot product means that the angle is obtuse.

(b) The scalar projection is

$$A_{\parallel} = \mathbf{A} \cdot \hat{b} = \frac{\mathbf{A} \cdot \mathbf{B}}{|\mathbf{B}|} = \frac{-6}{3} = -2,$$

and the vector projection is

$$A_{\parallel}\hat{b} = -2 \cdot \frac{1}{3}\left(\hat{\imath} - 2\hat{\jmath} + 2\hat{k}\right) = -\frac{2}{3}\hat{\imath} + \frac{4}{3}\hat{\jmath} - \frac{4}{3}\hat{k}.$$

(c)

$$\mathbf{A} \cdot \mathbf{C} = 2 + 3c = 0 \quad \Longrightarrow \quad c = -\frac{2}{3}.$$

(d)

$$W = (3)(2) + (4)(-1) = 6 - 4 = 2 \text{ J}.$$

Only the component of $\mathbf{F}$ along $\mathbf{d}$ does work.

(e)

$$|\mathbf{A} - \mathbf{B}|^2 = (\mathbf{A} - \mathbf{B}) \cdot (\mathbf{A} - \mathbf{B}) = A^2 + B^2 - 2\,\mathbf{A} \cdot \mathbf{B} = A^2 + B^2 - 2AB\cos\theta.$$

1.6.4 Example 4: Cross Product, Area, Torque and Volume

Let $\mathbf{A} = 2\hat{\imath} + \hat{\jmath} - \hat{k}$ and $\mathbf{B} = \hat{\imath} + 2\hat{\jmath} + \hat{k}$.

(a) Compute $\mathbf{A} \times \mathbf{B}$ and verify that $\mathbf{B} \times \mathbf{A} = -\mathbf{A} \times \mathbf{B}$.

(b) Verify that $\mathbf{A} \cdot (\mathbf{A} \times \mathbf{B}) = 0$ and find the unit normal $\hat{n}$ to the plane of $\mathbf{A}$ and $\mathbf{B}$.

(c) Find the area of the parallelogram and of the triangle spanned by $\mathbf{A}$ and $\mathbf{B}$. Obtain θ from $|\mathbf{A} \times \mathbf{B}|$ and check it with the dot product.

(d) For $\mathbf{r} = (2\hat{\imath} + \hat{\jmath})$ m and $\mathbf{F} = 3\hat{\jmath}$ N, find the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$.

(e) Find the volume of the parallelepiped with edges $\mathbf{a} = \hat{\imath} + 2\hat{\jmath}$, $\mathbf{b} = 3\hat{\jmath}$ and $\mathbf{c} = 4\hat{k}$.

Solution

(a)

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = \hat{\imath}(1+2) - \hat{\jmath}(2+1) + \hat{k}(4-1) = 3\hat{\imath} - 3\hat{\jmath} + 3\hat{k}.$$

Exchanging the two rows changes the sign, so

$$\mathbf{B} \times \mathbf{A} = -3\hat{\imath} + 3\hat{\jmath} - 3\hat{k}.$$

The cross product is anticommutative.

(b)

$$\mathbf{A} \cdot (\mathbf{A} \times \mathbf{B}) = 6 - 3 - 3 = 0,$$

so the result is perpendicular to both vectors. Since $|\mathbf{A} \times \mathbf{B}| = 3\sqrt{3}$,

$$\hat{n} = \frac{1}{\sqrt{3}} \left(\hat{\imath} - \hat{\jmath} + \hat{k} \right).$$

(c) The parallelogram area is $|\mathbf{A} \times \mathbf{B}| = 3\sqrt{3} \approx 5.20$ and the triangle area is $\frac{3\sqrt{3}}{2} \approx 2.60$. With $|\mathbf{A}| = |\mathbf{B}| = \sqrt{6}$,

$$\sin\theta = \frac{3\sqrt{3}}{6} = \frac{\sqrt{3}}{2}.$$

Checking with the dot product, $\mathbf{A} \cdot \mathbf{B} = 2 + 2 - 1 = 3$, so $\cos\theta = \frac{3}{6} = \frac{1}{2}$. Both give $\theta = 60^\circ$.

(d)

$$\boldsymbol{\tau} = (2\hat{\imath} + \hat{\jmath}) \times (3\hat{\jmath}) = 6(\hat{\imath} \times \hat{\jmath}) + 3(\hat{\jmath} \times \hat{\jmath}) = 6\hat{k} \text{ N} \cdot \text{m},$$

which points out of the page by the right-hand rule. The part of $\mathbf{r}$ parallel to $\mathbf{F}$ contributes nothing.

(e)

$$V = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})| = \left\| \begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{vmatrix} \right\| = 12.$$

1.6.5 Example 5: Derivatives, Kinematics, Maxima and Concavity

(a) Differentiate

$$f(x) = 3x^4 - 5x^2 + 7, \quad g(x) = x^2 \sin x, \quad h(x) = \sin(3x^2 + 1), \quad p(x) = \frac{x^2+1}{x-1}, \quad q(x) = e^{-x^2}.$$

(b) A ball is thrown so that $y(t) = 5 + 20t - 5t^2$, with y in metres and t in seconds. Find $v(t)$ and $a(t)$, the time and value of the maximum height, and confirm with the second derivative that it is a maximum.

(c) For simple harmonic motion $x(t) = A\sin(\omega t + \varphi)$, show that $a = -\omega^2 x$.

(d) For damped motion $x(t) = Ae^{-bt}\cos\omega t$, find $\dot{x}$.

Solution

(a) Using the power and sum rules, the product rule, the chain rule and the quotient rule,

$$\begin{aligned} f'(x) &= 12x^3 - 10x, \\ g'(x) &= 2x\sin x + x^2\cos x, \\ h'(x) &= 6x\cos(3x^2+1), \\ p'(x) &= \frac{2x(x-1) - (x^2+1)}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}, \\ q'(x) &= -2x\,e^{-x^2}. \end{aligned}$$

(b)

$$v = \dot{y} = 20 - 10t, \qquad a = \dot{v} = \ddot{y} = -10 \text{ m/s}^2.$$

The maximum occurs when $v = 0$, that is at $t = 2$ s, where

$$y(2) = 5 + 40 - 20 = 25 \text{ m}.$$

Since $y''(t) = -10 < 0$ everywhere, the curve is concave, so $t = 2$ s is indeed a maximum.

(c)

$$v = A\omega\cos(\omega t+\varphi), \qquad a = -A\omega^2\sin(\omega t+\varphi) = -\omega^2 x.$$

This is exactly the equation of motion produced by the spring force $F = -kx$.

(d) By the product and chain rules,

$$\dot{x} = A\left[-b\,e^{-bt}\cos\omega t - \omega\,e^{-bt}\sin\omega t\right] = -A\,e^{-bt}\left(b\cos\omega t + \omega\sin\omega t\right).$$

1.6.6 Example 6: Partial Derivatives, Total Differential and Total Derivative

(a) For $f(x,y) = x^2y + 3xy^3 - y$, find $\dfrac{\partial f}{\partial x}$ and $\dfrac{\partial f}{\partial y}$, and show that

$$\frac{\partial^2 f}{\partial x\,\partial y} = \frac{\partial^2 f}{\partial y\,\partial x}.$$

(b) The volume of a cylinder is $V = \pi r^2 h$. With $r = 2.0$ cm, $h = 10.0$ cm, $dr = 0.1$ cm and $dh = 0.2$ cm, estimate dV and the relative error dV/V.

(c) The temperature in a room is $T(x,y,t) = xy+5t$. A bug moves along $x(t) = t^2$ and $y(t) = 2t$. Find dT/dt in two different ways.

Solution

(a) Treating the other variable as a constant,

$$\frac{\partial f}{\partial x} = 2xy + 3y^3, \qquad \frac{\partial f}{\partial y} = x^2 + 9xy^2 - 1.$$

Then

$$\frac{\partial}{\partial y}\left(2xy+3y^3\right) = 2x + 9y^2, \qquad \frac{\partial}{\partial x}\left(x^2+9xy^2-1\right) = 2x + 9y^2,$$

so the mixed partial derivatives are equal.

(b)

$$dV = \frac{\partial V}{\partial r}dr + \frac{\partial V}{\partial h}dh = 2\pi rh\,dr + \pi r^2\,dh = 2\pi(2)(10)(0.1) + \pi(4)(0.2) = 4.8\pi \approx 15\ \text{cm}^3.$$

Since $V = \pi(4)(10) = 40\pi \approx 126\ \text{cm}^3$,

$$\frac{dV}{V} = \frac{4.8\pi}{40\pi} = 0.12 = 12\%, \qquad \text{or equivalently} \qquad \frac{dV}{V} = 2\frac{dr}{r} + \frac{dh}{h} = 2(0.05) + 0.02.$$

The exponent multiplies the relative error.

(c) Using the total derivative,

$$\frac{dT}{dt} = \frac{\partial T}{\partial x}\frac{dx}{dt} + \frac{\partial T}{\partial y}\frac{dy}{dt} + \frac{\partial T}{\partial t} = y(2t) + x(2) + 5 = (2t)(2t) + 2t^2 + 5 = 6t^2 + 5.$$

By direct substitution, $T = (t^2)(2t) + 5t = 2t^3 + 5t$, so

$$\frac{dT}{dt} = 6t^2 + 5,$$

which agrees. Physically, the temperature the bug feels changes both because it moves and because the room heats up, the latter being the term $\partial T/\partial t = 5$.

1.6.7 Example 7: Indefinite and Definite Integrals in Physics

(a) Evaluate

$$\int \left(3x^2 - 4x + 5\right) dx, \qquad \int \left(\frac{2}{x} + e^x\right) dx, \qquad \int \sin(2x)\, dx.$$

(b) Evaluate $\int_0^3 (2x+1)\, dx$ and check the answer geometrically using the area of a trapezoid. Then evaluate $\int_0^{\pi} \sin x\, dx$ and $\int_{-1}^{1} x^3\, dx$, and explain the second result.

(c) A particle has $a(t) = 6t$ m/s^2, with $v(0) = 2$ m/s and $x(0) = 0$. Find $v(t)$, $x(t)$, and the displacement during the first 2 s.

(d) Find the work done by a spring force $F = kx$ in stretching from 0 to x_0, with $k = 200$ N/m and $x_0 = 0.1$ m.

Solution

(a)

$$\int \left(3x^2 - 4x + 5\right) dx = x^3 - 2x^2 + 5x + C,$$

$$\int \left(\frac{2}{x} + e^x\right) dx = 2\ln|x| + e^x + C,$$

$$\int \sin(2x)\, dx = -\frac{1}{2}\cos 2x + C.$$

The last one can be checked by differentiating, since the chain rule returns $\sin 2x$.

(b)

$$\int_0^3 (2x+1)\, dx = \left[x^2 + x\right]_0^3 = 12.$$

Geometrically this is a trapezoid with parallel sides $f(0)=1$ and $f(3)=7$ and width 3, giving $\frac{1+7}{2}\cdot 3 = 12$. Furthermore,

$$\int_0^{\pi} \sin x\,dx = [-\cos x]_0^{\pi} = 1+1 = 2, \qquad \int_{-1}^{1} x^3\,dx = \left[\frac{x^4}{4}\right]_{-1}^{1} = \frac{1}{4}-\frac{1}{4} = 0.$$

The last integral vanishes because the negative area on $[-1,0]$ cancels the positive area on $[0,1]$; the definite integral gives a signed area.

(c)

$$v(t) = \int 6t\,dt = 3t^2 + C_1, \qquad v(0)=2 \implies C_1 = 2, \qquad v(t) = 3t^2+2,$$

$$x(t) = \int \left(3t^2+2\right) dt = t^3 + 2t + C_2, \qquad x(0)=0 \implies C_2 = 0,$$

$$\Delta x = \int_0^{2} v\,dt = \left[t^3+2t\right]_0^{2} = 8+4 = 12 \text{ m}.$$

The integration constants are fixed by the initial conditions, and this problem is the inverse of Example 5(b).

(d)

$$W = \int_0^{x_0} kx\,dx = \frac{1}{2}kx_0^2 = \frac{1}{2}(200)(0.1)^2 = 1 \text{ J}.$$

The force is not constant, so $W = Fd$ cannot be used; the integral gives the area under the F versus x line.

1.6.8 Example 8: Taylor Series and Physical Approximations

(a) Expand e^x and $\sin x$ about $x_0 = 0$ up to third order.

(b) Derive the binomial series $(1+x)^n \approx 1 + nx + \frac{n(n-1)}{2}x^2$ and use it to estimate $\sqrt{1.02}$.

(c) Show that the relativistic energy $E = mc^2/\sqrt{1-v^2/c^2}$ reduces to $mc^2 + \frac{1}{2}mv^2$ for $v \ll c$.

(d) Show that the gravitational potential energy $U = -GMm/(R+h)$ becomes a constant plus mgh for $h \ll R$.

Solution

(a) For $f(x) = e^x$ we have $f^{(n)}(0) = 1$ for every n, so

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots$$

For $f(x) = \sin x$ we have $f(0) = 0$, $f'(0) = 1$, $f''(0) = 0$ and $f'''(0) = -1$, so

$$\sin x = x - \frac{x^3}{3!} + \cdots \qquad \Longrightarrow \qquad \sin\theta \approx \theta,$$

which is the small-angle approximation used for the pendulum. Similarly $\cos x = 1 - \frac{x^2}{2} + \cdots$

(b) For $f(x) = (1+x)^n$ we have $f(0) = 1$, $f'(0) = n$ and $f''(0) = n(n-1)$, hence

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + \cdots \qquad \text{for } |x| \ll 1.$$

With $n = \frac{1}{2}$ and $x = 0.02$,

$$\sqrt{1.02} \approx 1 + 0.01 - \frac{1}{8}(0.0004) = 1.00995,$$

while the exact value is $1.009950\ldots$

(c) Setting $x = -v^2/c^2$ and $n = -\frac{1}{2}$,

$$E = mc^2\left(1 - \frac{v^2}{c^2}\right)^{-1/2} \approx mc^2\left(1 + \frac{1}{2}\frac{v^2}{c^2} + \frac{3}{8}\frac{v^4}{c^4}\right) = mc^2 + \frac{1}{2}mv^2 + \cdots$$

The first term is the rest energy and the second is the Newtonian kinetic energy.

(d)

$$U = -\frac{GMm}{R\,(1+h/R)} = -\frac{GMm}{R}\left(1 + \frac{h}{R}\right)^{-1} \approx -\frac{GMm}{R}\left(1 - \frac{h}{R}\right) = -\frac{GMm}{R} + \frac{GM}{R^2}mh,$$

and since $g = GM/R^2$ we obtain $U \approx \text{constant} + mgh$.

Closing remark: every simple formula of introductory physics, such as mgh, $\frac{1}{2}mv^2$ and $T = 2\pi\sqrt{L/g}$, is the first non-trivial term of a Taylor series.

1.7 Homeworks

1. Compute following indefinite integrals

$$\int \frac{dx}{a+x} \text{ and } \int \left(3x^2\, e^{x^3} + \frac{4x}{1+x^2}\right) dx,$$

where a is a constant.

2. Compute the definite integral

$$\int_1^4 (2x+3)\,dx.$$

3. Find the derivative of

$$f(x) = \frac{x^3 \ln(x)}{e^{2x}}, \qquad (x > 0).$$

4. Find the local minimum and maximum values of the function

$$f(x) = x^3 - 6x^2 + 9x + 1$$

by using the first and second derivative tests.

5. Find the Taylor series of $\sin x$, $\cos x$, and e^x about $x = 0$ up to and including the term in x^5.

6. Compute the partial derivatives of the function

$$f(x, y) = x^2 y + e^{xy}$$

with respect to x and y and write down the total differential df.

7. The position of a particle is described by the following vector as a function of time t:

$$\mathbf{r}(t) = a\cos(t)\,\hat{\imath} + a\sin(t)\,\hat{\jmath} + ct\,\hat{k}$$

where a and c are constants with units of length and velocity, respectively.

   (a) What kind of trajectory does the particle follow?

   (b) Find the velocity, acceleration and speed of the particle as functions of time.

   (c) When you consider the trajectory of the particle again, are the directions of the velocity and acceleration vectors in the way you expect? Explain.

   (d) Calculate the projection of the acceleration vector onto the velocity vector as a function of time and give a physical interpretation of this result.

8. A flat rectangular surface lies in the xy-plane. The surface has sides of length $a = 0.20$ m along the x-axis and $b = 0.30$ m along the y-axis. A uniform water flow is described by the flux vector field:

$$\mathbf{F} = 200\,\hat{\imath} + 300\,\hat{\jmath} + 400\,\hat{k} \quad \left[\frac{\mathrm{L}}{\mathrm{m}^2 \cdot \mathrm{s}}\right]$$

   (a) Compute the area vector $\mathbf{A}$ of the surface (the vector normal to the surface with magnitude equal to the area of the surface).

   (b) Compute the volume of water per unit time (in liters per second) passing through the surface using:

$$Q = \mathbf{F} \cdot \mathbf{A}.$$

   (c) Discuss how the orientation of the surface affects the water flow through it.

9. A particle moves in a plane such that its position vector at time t is given by

$$\mathbf{r}(t) = at^2\,\hat{\imath} + bt^3\,\hat{\jmath}.$$

   where a and b are constants with appropriate units.

   (a) Find the velocity and acceleration vectors as functions of time.

   (b) Find the angle between the velocity and acceleration vectors at time $t = 1$.

10. Given the vectors

$$\mathbf{A} = 2\,\hat{\imath} - 3\,\hat{\jmath} + \hat{k}, \quad \mathbf{B} = -\hat{\imath} + 4\,\hat{\jmath} + 2\,\hat{k},$$

    compute $\mathbf{A} \times \mathbf{B}$.

Solutions

1. **Indefinite integrals.**

   (a) Compute

$$\int \frac{dx}{a+x}.$$

   Let $u = a + x$, so that $du = dx$. Then

$$\int \frac{dx}{a+x} = \int \frac{du}{u} = \ln|u| + C = \ln|a+x| + C.$$

(b) Compute

$$\int \left(3x^2 e^{x^3} + \frac{4x}{1+x^2}\right) dx.$$

We split the integral:

$$\int 3x^2 e^{x^3}\, dx + \int \frac{4x}{1+x^2}\, dx.$$

For the first term, set $u = x^3$, $du = 3x^2\, dx$, so

$$\int 3x^2 e^{x^3}\, dx = \int e^u\, du = e^u + C = e^{x^3} + C.$$

For the second term, set $v = 1 + x^2$, $dv = 2x\, dx$, so $4x\, dx = 2\, dv$. Then

$$\int \frac{4x}{1+x^2}\, dx = \int \frac{2\, dv}{v} = 2\ln|v| + C = 2\ln(1+x^2) + C,$$

since $1 + x^2 > 0$.

Therefore,

$$\int \left(3x^2 e^{x^3} + \frac{4x}{1+x^2}\right) dx = e^{x^3} + 2\ln(1+x^2) + C.$$

2. **Definite integral.**

Compute

$$\int_1^4 (2x+3)\, dx.$$

An antiderivative of $2x + 3$ is

$$\int (2x+3)\, dx = x^2 + 3x + C.$$

Evaluate at the limits:

$$\begin{aligned}\int_1^4 (2x+3)\, dx &= \big[x^2 + 3x\big]_1^4 \\ &= \big(4^2 + 3\cdot 4\big) - \big(1^2 + 3\cdot 1\big) \\ &= (16+12) - (1+3) \\ &= 28 - 4 \\ &= 24.\end{aligned}$$

Interpretation: this is the area under the straight line $2x + 3$ between $x = 1$ and $x = 4$.

3. **Derivative of a function.**

   Given
   $$f(x) = \frac{x^3 \ln(x)}{e^{2x}} = x^3 \ln(x)\, e^{-2x}, \qquad x > 0.$$

   We differentiate using the product rule on $x^3 \ln(x)$ and e^{-2x}, or equivalently use the product rule plus chain rule.

   First compute
   $$g(x) = x^3 \ln(x), \qquad h(x) = e^{-2x}, \qquad f(x) = g(x)h(x).$$

   Then
   $$g'(x) = x\left(x^3 \ln x\right) = 3x^2 \ln x + x^3 \cdot \frac{1}{x} = 3x^2 \ln x + x^2 = x^2(3 \ln x + 1).$$

   Also
   $$h'(x) = x\left(e^{-2x}\right) = (-2)e^{-2x}.$$

   Product rule:
   $$\begin{aligned} f'(x) &= g'(x)h(x) + g(x)h'(x) \\ &= \left[x^2(3 \ln x + 1)\right] e^{-2x} + \left[x^3 \ln x\right](-2e^{-2x}) \\ &= e^{-2x}\left[x^2(3 \ln x + 1) - 2x^3 \ln x\right]. \end{aligned}$$

   Factor x^2:
   $$f'(x) = x^2 e^{-2x}\left[(3 \ln x + 1) - 2x \ln x\right].$$

   So
   $$f'(x) = x^2 e^{-2x}\left(3 \ln x + 1 - 2x \ln x\right), \qquad x > 0.$$

   Explanation: we used product rule and the fact that $x(\ln x) = 1/x$ for $x > 0$.

4. **Local extrema via first and second derivative tests.**

   Let
   $$f(x) = x^3 - 6x^2 + 9x + 1.$$

   First derivative:
   $$f'(x) = 3x^2 - 12x + 9.$$

   Set $f'(x) = 0$:

   $$3x^2 - 12x + 9 = 0 \quad \Longrightarrow \quad x^2 - 4x + 3 = 0 \quad \Longrightarrow \quad (x-1)(x-3) = 0.$$

So critical points are $x = 1$ and $x = 3$.

Second derivative:

$$f''(x) = 6x - 12.$$

Evaluate f'':

$$f''(1) = 6(1) - 12 = -6 < 0,$$

which means f is concave down at $x = 1$, so $x = 1$ is a local maximum.

$$f''(3) = 6(3) - 12 = 6 > 0,$$

which means f is concave up at $x = 3$, so $x = 3$ is a local minimum.

Now compute the function values:

$$f(1) = 1^3 - 6(1)^2 + 9(1) + 1 = 1 - 6 + 9 + 1 = 5,$$
$$f(3) = 3^3 - 6(3)^2 + 9(3) + 1 = 27 - 54 + 27 + 1 = 1.$$

Conclusion:

local maximum: $f(1) = 5$ at $x = 1$, local minimum: $f(3) = 1$ at $x = 3$.

Explanation: the sign of f'' at the critical point determines if the point is a max ($f'' < 0$) or a min ($f'' > 0$).

5. **Taylor series of $\sin x$, $\cos x$, e^x about 0 up to x^5.**

Maclaurin series (Taylor about 0):

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \mathcal{O}(x^7) = x - \frac{x^3}{6} + \frac{x^5}{120} + \mathcal{O}(x^7).$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \mathcal{O}(x^6) = 1 - \frac{x^2}{2} + \frac{x^4}{24} + \mathcal{O}(x^6).$$

$$e^x = 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\frac{x^4}{4!}+\frac{x^5}{5!}+\mathcal{O}(x^6) = 1+x+\frac{x^2}{2}+\frac{x^3}{6}+\frac{x^4}{24}+\frac{x^5}{120}+\mathcal{O}(x^6).$$

Explanation: we truncated after the x^5 term or the highest power ≤ 5 that actually appears.

6. **Partial derivatives and total differential.**

   Given
   $$f(x, y) = x^2 y + e^{xy}.$$

   Partial derivative with respect to x:

   $$f_x = fx = 2xy + e^{xy} \cdot x(xy) = 2xy + e^{xy} \cdot y = 2xy + ye^{xy}.$$

   Partial derivative with respect to y:

   $$f_y = fy = x^2 + e^{xy} \cdot y(xy) = x^2 + e^{xy} \cdot x = x^2 + xe^{xy}.$$

   The total differential is

   $$df = f_x\, dx + f_y\, dy = (2xy + ye^{xy})\, dx + (x^2 + xe^{xy})\, dy.$$

   Explanation: df gives the linear change in f under small changes dx, dy.

7. **Particle motion on a helix.**

   The position is
   $$\mathbf{r}(t) = a\cos t\, \hat{\imath} + a \sin t\, \hat{\jmath} + ct\, \hat{k},$$
   where a has units of length and c has units of velocity.

   (a) *Type of trajectory.*
   The x-y coordinates satisfy

   $$x(t) = a\cos t, \qquad y(t) = a \sin t.$$

   Then
   $$x^2 + y^2 = a^2(\cos^2 t + \sin^2 t) = a^2,$$
   which is a circle of radius a in the xy-plane. Meanwhile $z(t) = ct$ increases linearly in time. Therefore the path is a helix of radius a rising at constant rate c along the z-axis.

   (b) *Velocity, acceleration, and speed.*
   Velocity:
   $$\mathbf{v}(t) = \mathbf{r}t = -a \sin t\, \hat{\imath} + a\cos t\, \hat{\jmath} + c\, \hat{k}.$$

   Acceleration:

   $$\mathbf{a}(t) = \mathbf{v}t = -a\cos t\, \hat{\imath} - a \sin t\, \hat{\jmath} + 0\, \hat{k} = -a\cos t\, \hat{\imath} - a \sin t\, \hat{\jmath}.$$

Speed (magnitude of velocity):

$$\begin{aligned} v(t) = \|\mathbf{v}(t)\| &= \sqrt{(-a\sin t)^2 + (a\cos t)^2 + c^2} \\ &= \sqrt{a^2(\sin^2 t + \cos^2 t) + c^2} \\ &= \sqrt{a^2 + c^2}. \end{aligned}$$

The speed is constant in time.

(c) *Directions of* $\mathbf{v}$ *and* $\mathbf{a}$.
In the xy-plane, $(-a\sin t, a\cos t)$ is tangent to the circle $x^2+y^2 = a^2$, and $(-a\cos t, -a\sin t)$ points radially inward toward the center $(0,0)$.
So:

$\mathbf{v}(t)$ is tangent to the circular motion in the xy-plane (plus upward motion in $\hat{k}$),

while

$\mathbf{a}(t)$ points toward the axis of the circle in the xy-plane (centripetal), with no $\hat{k}$ c

This matches uniform circular motion of radius a in the horizontal projection: velocity is tangent, acceleration is inward.

(d) *Projection of* $\mathbf{a}$ *onto* $\mathbf{v}$. *Physical meaning.*
The projection of $\mathbf{a}$ onto $\mathbf{v}$ is the component of acceleration along the direction of motion. It measures tangential acceleration (change in speed). Compute

$$\mathrm{proj}_{\mathbf{v}}\mathbf{a} = \frac{\mathbf{a}\cdot\mathbf{v}}{\|\mathbf{v}\|^2}\,\mathbf{v}.$$

First compute the dot product:

$$\begin{aligned} \mathbf{a}(t)\cdot\mathbf{v}(t) &= (-a\cos t)(-a\sin t) + (-a\sin t)(a\cos t) + (0)(c) \\ &= a^2\cos t\sin t - a^2\sin t\cos t \\ &= 0. \end{aligned}$$

Therefore

$$\mathrm{proj}_{\mathbf{v}}\mathbf{a} = \mathbf{0}.$$

Interpretation: there is no acceleration component along the velocity direction, so the speed is constant. The acceleration is purely normal (centripetal), changing only the direction of the velocity, not its magnitude.

8. **Flux through a surface in a vector field.**

Given a flat rectangle in the xy-plane with side lengths

$$a = 0.20 \text{ m} \quad (\text{along } x), \qquad b = 0.30 \text{ m} \quad (\text{along } y),$$

and flux density field

$$\mathbf{F} = 200\,\hat{\imath} + 300\,\hat{\jmath} + 400\,\hat{k} \quad \left[\frac{\text{L}}{\text{m}^2 \cdot \text{s}}\right].$$

(a) *Area vector* $\mathbf{A}$.

$$\mathbf{A} = (a\hat{\imath}) \times (b\hat{\jmath}) = (0.20 \text{ m})(0.30 \text{ m})\hat{k} = (0.060)\hat{k} \quad [\text{m}^2].$$

Its normal vector is in the $+\hat{k}$ direction perpendicular to the surface lying in the xy-plane.

(b) *Flow rate* $Q = \mathbf{F} \cdot \mathbf{A}$.
Compute the dot product:

$$\begin{aligned} Q &= \mathbf{F} \cdot \mathbf{A} \\ &= (200\,\hat{\imath} + 300\,\hat{\jmath} + 400\,\hat{k}) \cdot (0.060\,\hat{k}) \\ &= 400 \cdot 0.060 \quad \left[\frac{\text{L}}{\text{m}^2 \cdot \text{s}}\right] \cdot [\text{m}^2] \\ &= 24 \text{ L/s}. \end{aligned}$$

Units: $\mathbf{F}$ is volume flux per area per time, $\mathbf{A}$ is area, so the product is volume per time.

(c) *Effect of orientation.*
In general,

$$Q = \mathbf{F} \cdot \mathbf{A} = |\mathbf{F}|\,|\mathbf{A}| \cos\theta,$$

where θ is the angle between $\mathbf{F}$ and the surface normal. If you tilt the surface, its area vector $\mathbf{A}$ tilts. The effective flux is reduced by $\cos\theta$.

- If the surface is perpendicular to the flow (normal aligned with $\mathbf{F}$), $\theta = 0$, $\cos\theta = 1$, and Q is maximal.
- If the surface is parallel to the flow (normal perpendicular to $\mathbf{F}$), $\theta = 90°$, $\cos\theta = 0$, and $Q = 0$.

Here the surface normal is $\hat{k}$, so only the k-component of $\mathbf{F}$ contributes to the flux.

9. **Velocity, acceleration, and their angle.**

Given

$$\mathbf{r}(t) = at^2\,\hat{\imath} + bt^3\,\hat{\jmath}.$$

(a) *Velocity and acceleration.*

Velocity:

$$\mathbf{v}(t) = \mathbf{r}t = 2at\,\hat{\imath} + 3bt^2\,\hat{\jmath}.$$

Acceleration:

$$\mathbf{a}(t) = \mathbf{v}t = 2a\,\hat{\imath} + 6bt\,\hat{\jmath}.$$

(b) *Angle between* $\mathbf{v}$ *and* $\mathbf{a}$ *at* $t = 1$.

At $t = 1$:

$$\mathbf{v}(1) = 2a\,\hat{\imath} + 3b\,\hat{\jmath}, \qquad \mathbf{a}(1) = 2a\,\hat{\imath} + 6b\,\hat{\jmath}.$$

The angle θ between two vectors $\mathbf{u}$ and $\mathbf{w}$ is given by

$$\cos\theta = \frac{\mathbf{u}\cdot\mathbf{w}}{\|\mathbf{u}\|\,\|\mathbf{w}\|}.$$

Compute the dot product:

$$\mathbf{v}(1)\cdot\mathbf{a}(1) = (2a)(2a) + (3b)(6b) = 4a^2 + 18b^2.$$

Compute magnitudes:

$$\|\mathbf{v}(1)\| = \sqrt{(2a)^2 + (3b)^2} = \sqrt{4a^2 + 9b^2},$$

$$\|\mathbf{a}(1)\| = \sqrt{(2a)^2 + (6b)^2} = \sqrt{4a^2 + 36b^2}.$$

Therefore

$$\cos\theta = \frac{4a^2 + 18b^2}{\sqrt{4a^2 + 9b^2}\,\sqrt{4a^2 + 36b^2}}.$$

So the angle θ is

$$\theta = \arccos\left(\frac{4a^2 + 18b^2}{\sqrt{4a^2 + 9b^2}\,\sqrt{4a^2 + 36b^2}}\right).$$

Interpretation: the angle depends on a and b; if $\mathbf{a}$ is parallel to $\mathbf{v}$, then the direction of motion is speeding up with no change in direction. If not parallel, part of $\mathbf{a}$ turns the velocity.

10. **Cross product.**

Given

$$\mathbf{A} = 2\,\hat{\imath} - 3\,\hat{\jmath} + 1\,\hat{k}, \qquad \mathbf{B} = -1\,\hat{\imath} + 4\,\hat{\jmath} + 2\,\hat{k}.$$

The cross product $\mathbf{A} \times \mathbf{B}$ is

$$\begin{aligned}
\mathbf{A} \times \mathbf{B} &= (2\hat{\imath} - 3\hat{\jmath} + \hat{k}) \times (-\hat{\imath} + 4\hat{\jmath} + 2\hat{k}) \\
&= 2\hat{\imath} \times (-\hat{\imath}) + 2\hat{\imath} \times 4\hat{\jmath} + 2\hat{\imath} \times 2\hat{k} - 3\hat{\jmath} \times (-\hat{\imath}) - 3\hat{\jmath} \times 4\hat{\jmath} - 3\hat{\jmath} \times 2\hat{k} \\
&\quad + \hat{k} \times (-\hat{\imath}) + \hat{k} \times 4\hat{\jmath} + \hat{k} \times 2\hat{k} \\
&= 0 + 8\hat{k} - 4\hat{\jmath} - 3\hat{k} + 0 - 6\hat{\imath} - \hat{\jmath} - 4\hat{\imath} + 0 \\
&= -10\,\hat{\imath} - 5\,\hat{\jmath} + 5\,\hat{k}.
\end{aligned}$$

Explanation: $\mathbf{A} \times \mathbf{B}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$ and its magnitude is $|\mathbf{A}||\mathbf{B}| \sin\phi$, where ϕ is the angle between them.